FitTrack Product Analytics & Growth Strategy

Full-Stack Analytics Implementation: RFM Segmentation, A/B Power Analysis & Kaplan-Meier Survival Modeling

1 & 2. Business Scenario & Operational Friction Points

FitTrack is a fast-growing mobile fitness application preparing for its Series B fundraising round. As user acquisition scaled from 100K to 500K Monthly Active Users (MAU), the company faced critical alignment challenges across marketing acquisition channels and long-term user retention. Leadership required a robust, data-grounded analytics pipeline to optimize a **$2.0 Million marketing budget** and prove sustainable unit economics to investors.

Pain Point	Operational Impact	Analytical Solution
Siloed Channel Analytics	Marketing optimized top-of-funnel signups without evaluating downstream user retention or LTV.	Unified RFM & cohort retention metrics.
Unverified Channel Quality	High initial social media traffic masked severe 8-week churn (82% week 1 down to 8% week 8).	Kaplan-Meier survival modeling in R.
Novelty Bias in A/B Testing	Early 12% treatment conversion spikes caused premature rollouts of features users abandoned after 3 days.	Mandated 14-day test runtime protocols.
Arbitrary Sample Sizing	Lack of power analysis led to underpowered split tests and unreliable feature decisions.	Calculated power benchmark (n = 9,604/grp).

3. Project Goals & Tool Rationale

To solve these friction points, we built an integrated three-pillar analytics methodology: **RFM K-Means Segmentation**, **A/B Power Analysis**, and **Kaplan-Meier Survival Analysis**.

Analytics Framework	Why We Use It (2-3 Lines)	Business Impact & Value Created
RFM K-Means Segmentation	Evaluates Recency (days active), Frequency (sessions), and Monetary spend to group 10,000 users into 4 clear behavioral profiles.	Stops ad spend waste on dormant accounts and directs VIP incentives to keep high-value regulars active.
A/B Power Analysis	Calculates required sample size (n = 9,604/group) to guarantee split test results reflect true 2% product lifts at 80% power.	Prevents multi-million-dollar feature rollouts driven by temporary 3-day novelty curiosity spikes.

Kaplan-Meier Survival Analysis	Tracks customer retention 'lifespans' over time in R and proves statistical significance via Log-Rank test (Chi-Square = 7,730.51, p < 0.0001).	Identifies exact churn drop-off windows (Days 14-30) and justifies reallocating $2M toward high-LTV Referral channels.

Technical Pipeline Architecture (Google Colab Execution)

1. generate_data.py: Created synthetic benchmark data matching 10,000 users, 21-day experiment logs, and 12-week retention cohorts across 5 channels.

2. product_analytics.py: Executed data imputation ($0.00 revenue), RFM feature scaling, K-Means clustering (k=4), and sample size calculation.

3. survival_analysis.R: Fitted Kaplan-Meier survival curves, computed Log-Rank test statistics, and exported publication-quality visualizations.

4. Statistical Data Findings & Quantitative Observations

Statistical Metric / Value	Plain-English Meaning	Business Value & Revenue Impact
k = 4 Clusters	Divided user base into High-Value Regulars, At-Risk Champions, Engaged Low-Spenders, and Casual/Dormant.	Reveals 20% of users drive >60% revenue; flags At-Risk Champions before they churn.
n = 9,604 / group	Required sample size for 80% statistical power and 2% effect size at alpha=0.05.	Ensures experiment decisions are grounded in statistically reliable user sample sizes.
12% vs 7% Novelty Spike	Treatment conversion jumped to 12% in days 1-3 due to curiosity, but leveled off to 8% afterward.	Saves engineering costs by preventing rollouts of features users abandon after 3 days.
Chi-Square = 7,730.51 (p < 0.0001)	Log-Rank test proves retention curves across user segments are fundamentally different (0% random chance).	Proves user segments have distinct lifespans, providing mathematical proof for marketing budget changes.

5. Executive Recommendations & Strategic Roadmap

Recommendation 1: Reallocate $2.0M Ad Budget — Shift 65% of marketing funds to Referral and Email acquisition channels (sustaining 45% 8-week retention) while cutting Social Media spend from $600K to $300K.

Recommendation 2: Enforce 14-Day A/B Testing Protocols — Mandate minimum 14-day test runtimes and n = 9,604 per group sample sizes to eliminate novelty bias before product feature approvals.

Recommendation 3: Automate At-Risk Champion Win-Back Triggers — Deploy targeted push/email re-engagement offers between Day 14 and Day 30 when At-Risk Champions show activity decay.

Recommendation 4: Present Unit Economics in Series B Deck — Feature Kaplan-Meier survival curves and segment LTV metrics in investor pitch materials to demonstrate predictable, scalable growth.